Quadratics & Nonlinear Equations

ANSWER KEY

Section 1: Factoring & Zero Product Property

1 What is the sum of the solutions to the quadratic equation $2x^2 - 11x + 12 = 0$?

a. $\frac{11}{2}$ ✓

b. $\frac{12}{2}$

c. $-\frac{11}{2}$

d. 6

By Vieta's formulas, the sum of the roots of $ax^2 + bx + c = 0$ is $-\frac{b}{a} = -\frac{-11}{2} = \frac{11}{2}$.

2 If $(2x - 3)(x + 5) = 0$, what is the positive solution to the equation?

a. $\frac{2}{3}$

b. $\frac{3}{2}$ ✓

c. 5

d. -5

Setting factors to zero: $2x - 3 = 0 \Rightarrow x = \frac{3}{2}$, or $x + 5 = 0 \Rightarrow x = -5$. The positive root is $\frac{3}{2}$.

3 If $x^2 - 16 = (x - c)(x + c)$ for all real numbers x, where $c > 0$, what is the value of c?

a. 2

b. 4 ✓

c. 8

d. 16

Difference of squares: $x^2 - 16 = x^2 - c^2 \Rightarrow c^2 = 16 \Rightarrow c = 4$.

Section 2: Vertex Form & Minima/Maxima

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4 A parabola in the xy-plane has equation $y = 3(x - 4)^2 - 7$. What are the coordinates of its vertex?

a. $(-4, -7)$

b. $(4, -7)$ ✓

c. $(4, 7)$

d. $(-4, 7)$

In vertex form $y = a(x - h)^2 + k$, the vertex is (h, k). Here $h = 4$ and $k = -7$, so the vertex is $(4, -7)$.

5 What is the minimum value of the quadratic function $f(x) = x^2 - 6x + 14$?

a. 3

b. 5 ✓

c. 9

d. 14

Complete the square: $f(x) = (x^2 - 6x + 9) + 5 = (x - 3)^2 + 5$. Since $(x - 3)^2 \geq 0$, the minimum value is 5.

6 The height h in feet of a projectile t seconds after launch is given by $h(t) = -16t^2 + 64t + 80$. What is the maximum height, in feet, reached by the projectile?

a. 120

b. 144 ✓

c. 160

d. 184

The vertex occurs at $t = -\frac{b}{2a} = -\frac{64}{2(-16)} = 2$ seconds. Then $h(2) = -16(4) + 64(2) + 80 = -64 + 128 + 80 = 144$ feet.

Section 3: Discriminant Analysis ($b^2 - 4ac$)

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7 For what value of c does the equation $x^2 - 8x + c = 0$ have exactly one real solution?

a. 8

b. 16 ✓

c. 32

d. 64

Exactly one real solution means the discriminant equals zero:
$b^2 - 4ac = 0 \Rightarrow (-8)^2 - 4(1)(c) = 0 \Rightarrow 64 - 4c = 0 \Rightarrow c = 16$.

8 Which of the following quadratic equations has no real solutions?

a. $x^2 - 4x + 3 = 0$

b. $x^2 - 2x + 1 = 0$

c. $x^2 - 2x + 5 = 0$ ✓

d. $x^2 + 3x - 4 = 0$

Calculate discriminant $\Delta = b^2 - 4ac$: for $x^2 - 2x + 5 = 0$, $\Delta = (-2)^2 - 4(1)(5) = 4 - 20 = -16 < 0$, so it has no real solutions.

9 If the quadratic equation $3x^2 - kx + 12 = 0$ has two distinct real roots, which condition must k satisfy?

a. $k^2 > 144$ ✓

b. $k^2 < 144$

c. $k^2 = 144$

d. $k > 12$

Two distinct real roots require $\Delta > 0 \Rightarrow (-k)^2 - 4(3)(12) > 0 \Rightarrow k^2 - 144 > 0 \Rightarrow k^2 > 144$.

Section 4: Nonlinear Systems & Intersections

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10 How many points of intersection exist between the line $y = 2x - 3$ and the parabola $y = x^2 - 4x + 6$?

a. 0

b. 1 ✓

c. 2

d. 3

Set $x^2 - 4x + 6 = 2x - 3 \Rightarrow x^2 - 6x + 9 = 0 \Rightarrow (x - 3)^2 = 0$. Since there is a single repeated root $x = 3$, there is exactly 1 point of intersection.

11 If (x, y) is a solution to the system $y = x^2 + 3$ and $y = 4x$, what is the sum of all possible values of x?

a. 3

b. 4 ✓

c. 7

d. 12

Set $x^2 + 3 = 4x \Rightarrow x^2 - 4x + 3 = 0 \Rightarrow (x - 1)(x - 3) = 0$. Roots are $x = 1$ and $x = 3$. Their sum is $1 + 3 = 4$.

12 A circle in the xy-plane has equation $(x - 3)^2 + (y + 2)^2 = 25$. At what points does the line $x = 3$ intersect the circle?

a. $(3, 3)$ **and** $(3, -7)$ ✓

b. $(3, 5)$ and $(3, -5)$

c. $(0, 3)$ and $(0, -7)$

d. $(3, 2)$ and $(3, -2)$

Substitute $x = 3$: $(3 - 3)^2 + (y + 2)^2 = 25 \Rightarrow (y + 2)^2 = 25 \Rightarrow y + 2 = \pm 5$. Thus $y = 3$ or $y = -7$. The points are $(3, 3)$ and $(3, -7)$.

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